Warm Starts, Honest Clocks: Dual-Horizon Radar Alignment for Distributed Replay

Extended research manuscript

Abstract

Distributed automotive radar replay must initialize calibration state as well as object tracks. We formulate an eligibility-indexed dual-horizon alignment schedule and distinguish causal cold-start adaptation from provenance-qualified historical seeding. Exact bias and variance expressions expose a finite startup-gain budget, while yaw-invariant range–rate relations separate timestamp offset from radial-speed bias. Reproducing a ten-seed scalar experiment gives 46.2 frames to first tolerance entry; a 100-seed extension averages 41.14 frames. Two alternative gain settings remain censored at 400 frames under the same protocol, demonstrating that a fast-lock claim cannot be transferred between configurations. Controlled timing fixtures attain 6.632 and 3.799 ms RMSE in radial and tangential geometries. The study provides a reproducible initialization argument rather than inferring cluster throughput from estimator delay.

1 Introduction

The first vehicle in a replay shard is rarely the first vehicle the original radar saw. By the time a continuous drive reaches a shard boundary, the sensor has already learned a mounting correction, accumulated clutter statistics, and established tracks. A worker starting from an empty state inherits none of that history. The resulting error can look like a tracking failure even when it originates much earlier, in the calibration that rotates every detection entering the tracker. This distinction matters when a long drive is divided among compute workers. A short preceding segment may be sufficient to initialize position and velocity, yet insufficient to recover a calibration state that updates only during suitable motion. Adding the same overlap to every worker is therefore a convenient scheduling rule, not a convergence argument. The relevant horizon is the number and quality of informative updates. We study two complementary answers. During a genuine cold start, an eligibility-indexed gain allocates more adaptation to the beginning of the sequence and then returns to a conservative steady regime. During historical replay, a valid recorded state can initialize the worker directly. These answers must remain separate: a retrospective state is useful for reproducing a recording, but it would leak future information into an online cold-start experiment. The paper makes three contributions. First, it derives a calibration/replay contract that distinguishes elapsed frames, accepted updates and state provenance. Second, it provides finite-horizon error and noise analysis for a dual-horizon recursion, together with a yaw-independent clock-bias formulation for aligning asynchronous observations. Third, it revisits the experimental claims under explicit configurations. Reproducing the original ten-seed race yields 46.2 frames to first tolerance entry for one gain setting; applying two other, previously conflated settings does not reach that tolerance within the same 400-frame experiment. The difference is central to the paper: fast initialization is a property of a complete estimator configuration, not a formula printed beside a desirable number.

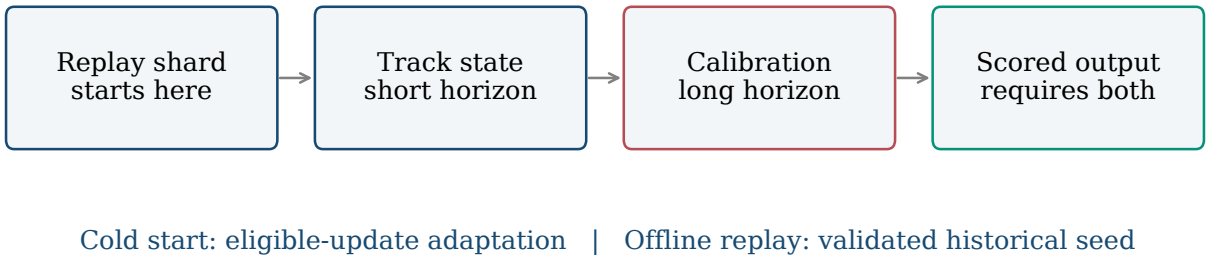


Figure 1: Replay initialization has two horizons. Scoring is valid only after the state required by the entire perception path is available. Historical seeding and cold-start adaptation solve different operating cases.

2 Related Work and Positioning

Recursive state estimation provides the mathematical foundation [1]; it does not prescribe a universal warm-up duration. Calibration adds a second source of state whose observability depends on scene geometry and vehicle motion. Our concern is the interface between this state and distributed replay, rather than a claim that exponential gain schedules are new in isolation. Temporal parallelization of Bayesian filtering and smoothing replaces sequential recursions by associative operators under suitable models [7]. That addresses computational dependency inside an estimator. It does not automatically restore the state of a nonlinear production front end, nor preserve decisions made by thresholded association logic. We therefore treat replay initialization and temporal parallel filtering as complementary problems. The timing analysis uses rotation-invariant inner products to separate clock offset from yaw under a stated kinematic model. The residual beat estimator is an application of phase-increment frequency estimation, whose established lineage includes Kay [3]. We include it as a resource-aware supporting component, not as the central novelty of calibration initialization. The distinction keeps the main argument focused: replay should either reconstruct a valid state or establish why its initialization error cannot change the scored decisions.

3 Problem Formulation

Let $\theta^\star$ be a locally constant boresight angle and $\hat{\theta}_k$ its estimate after elapsed frame k. A binary eligibility variable a_k indicates whether that frame supplies a trustworthy calibration observation. The accepted-update counter is

$$N_k = \sum_{i=0}^{k} a_i. \tag{1}$$

Let $z_k = \theta^\star + \eta_k$ denote a local angle observation. An explicit two-regime recursion is

$$\hat{\theta}_{k+1} = \hat{\theta}_k + g_k(z_k - \hat{\theta}_k), \tag{2}$$

$$g_k = K_s + a_k K_f \exp(-\alpha N_k). \tag{3}$$

The gains K_s, K_f are dimensionless and α is per accepted update. When $a_k = 0$, this model still applies the slow update. A strict rejection policy would instead set $g_k = 0$; that is a different estimator and must not be substituted silently in a comparison. The observation model is a local reduction of a calibration problem,

not an identification of an inaccessible production filter. It is useful because it exposes the effect of the startup schedule without hiding it behind a large software stack. The analysis assumes a fixed angle over the evaluated horizon, a documented sensor/ego coordinate convention, and a declared policy for rejected observations.

3.1 What does "lock" mean?

We distinguish first tolerance entry,

$$T_{\text{first}} = \inf\{k+1 : |\hat{\theta}_k - \theta^\star| < \delta\}, \tag{4}$$

from persistent lock over H successive frames. First entry is easier to achieve under noise and says nothing about a later departure. An estimate-change test, such as $|\hat{\theta}_k - \hat{\theta}_{k-1}| < \delta$, measures motion of the estimator rather than error relative to the true mounting angle. A frozen but wrong estimate passes that test. All experimental lock results below refer to first entry unless explicitly stated otherwise.

4 Dual-Horizon Analysis

4.1 Bias contraction

With $e_k = \hat{\theta}_k - \theta^\star$, the noiseless error satisfies

$$e_m = e_0 \prod_{k=0}^{m-1} (1 - g_k). \tag{5}$$

For $0 < g_k < 1$, its magnitude contracts monotonically. Under independent zero-mean observation noise, the same product governs the mean error. In the special case of exclusively eligible observations, $N_k = k+1$ and

$$\frac{|e_m|}{|e_0|} \le \exp\left[-mK_s - K_f \frac{e^{-\alpha}(1 - e^{-m\alpha})}{1 - e^{-\alpha}}\right]. \tag{6}$$

This follows from $\log(1-u) \le -u$. It provides a sufficient contraction bound; it is not an exact lock-time expression in the presence of noise. The finite sum also explains a subtle limitation. The fast component supplies a finite total adaptation budget, whereas the slow component accumulates indefinitely. If the initial fast budget is too small, a schedule can decay almost completely before the estimate reaches the desired tolerance. The final stage then reverts to the same slow convergence one was trying to avoid.

4.2 Noise cost of moving quickly

If the eligibility sequence is fixed and η_k is independent with variance σ_k^2, then

$$P_{k+1} = (1 - g_k)^2 P_k + g_k^2 \sigma_k^2. \tag{7}$$

The exact finite-horizon expression separates transported initialization uncertainty from injected measurement uncertainty:

$$\begin{aligned} P_m = P_0 \prod_{i=0}^{m-1} (1 - g_i)^2 \\ + \sum_{j=0}^{m-1} g_j^2 \sigma_j^2 \prod_{i=j+1}^{m-1} (1 - g_i)^2. \end{aligned} \tag{8}$$

For a fixed slow gain and constant variance, $P_\infty = K_s \sigma^2 / (2 - K_s)$. A larger startup gain suppresses initialization bias but admits more early measurement noise. The stationary-scatterer gate is therefore part of the estimator, not a decorative safeguard added after tuning.

4.3 Physical qualification

For a stationary world point, let $\mathbf{u}$ be the sensor-to-point unit vector and $\mathbf{v}_s$ the sensor velocity in the same frame. A Doppler consistency test is

$$|v_r + \mathbf{u}^T \mathbf{v}_s| \leq \varepsilon_v. \tag{9}$$

Lever-arm rotation contributes to $\mathbf{v}_s$ and cannot be ignored on a turning vehicle. A Doppler alias must be resolved before applying this test. The gate admits stationary guardrails when they satisfy the model; it does not reject them merely because they are infrastructure. Independent geometric diversity and robust angle fitting are still needed to prevent a poorly observed scene from producing a confident mounting correction.

5 Preflight Seeding Without Look-Ahead Confusion

Historical replay offers information that an online cold start does not. A recorded alignment state can be attached to a worker's initial condition, provided its meaning is explicit. Denote the seed by $(\theta_0, P_0, \mathscr{P})$, where $\mathscr{P}$ records sensor identity, mounting configuration, epoch and quality. Acceptance requires compatibility with the replayed sensor and a documented relationship between the state epoch and the scored segment. A mounting-angle range check catches gross corruption but does not establish convergence. Finite-value checks, age, configuration consistency and uncertainty must be considered separately. If only a point estimate is available, treating it as exact silently understates uncertainty. Where a covariance is available, the seeded error is propagated using (7) rather than reset to zero by assertion. An extracted example in the project records azimuth -1.145°. That is a concrete transport example, not an accuracy guarantee for every worker. The HDF extraction prototype averages a tail window and uses record length as a convergence proxy; a production criterion should consume the recorded quality and uncertainty channels directly. Offline reconstruction and online evaluation must use distinct modes so a future-converged value cannot inadvertently improve a claimed online cold-start result.

6 Temporal Alignment of Asynchronous Sensors

Calibration consistency is incomplete if sensors are fused at different epochs. Under the wrapper's stated ego-frame convention, a second-order correction is

$$\mathbf{p}_c = R(\omega \Delta t)\mathbf{p} - \mathbf{v}_e \Delta t - \tfrac{1}{2}\mathbf{a}_e \Delta t^2. \tag{10}$$

Position, velocity, acceleration and yaw rate use metres, m/s, m/s^2 and rad/s, respectively. The sign of Δt and the direction of the frame map define the rotation sign; the formula is not transferable between coordinate conventions without that definition.

6.1 Clock offset before yaw

Let an independent track predict $\mathbf{d}$ and velocity $\mathbf{v}$ at a reference epoch. For a constant-velocity offset τ, the contemporaneous point is $\mathbf{x} = \mathbf{d} + \tau\mathbf{v}$. Its range and radial velocity satisfy

$$rw - \mathbf{d}^T \mathbf{v} = \|\mathbf{v}\|^2 \tau. \tag{11}$$

A common yaw rotation does not change the inner product. With additive radial-speed bias b, define $y_i = r_i w_i - \mathbf{d}_i^T \mathbf{v}_i$ and $a_i = \|\mathbf{v}_i\|^2$. Two targets give

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_1 & r_1 \\ a_2 & r_2 \end{bmatrix} \begin{bmatrix} \tau \\ b \end{bmatrix}. \tag{12}$$

Table 1: Alignment configurations; first-entry horizon 400 frames

Setting	K_f	K_s	α	Censored
F1	0.15	0.004	0.05	0/100
F2	0.05	0.0025	0.08	100/100
F3	0.05	0.0025	0.1	100/100

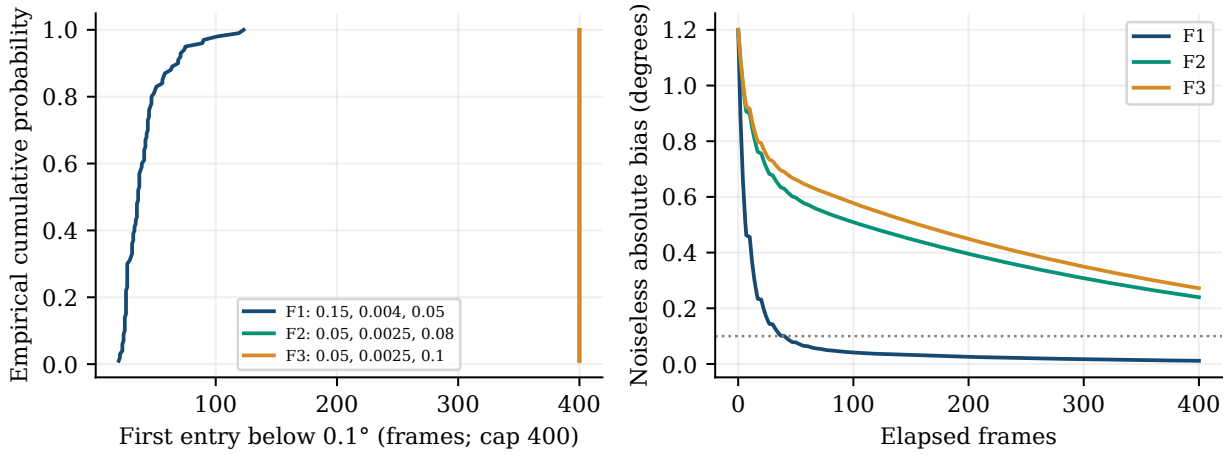


Figure 2: Configuration-controlled alignment study. Left: first-entry distributions over 100 seeds; a mass at 400 denotes censoring. Right: exact noiseless bias for the same eligibility pattern. F1 uses $(K_f, K_s, \alpha) = (0.15, 0.004, 0.05)$; F2/F3 use $(0.05, 0.0025, 0.08/0.10)$.

The determinant $a_1 r_2 - a_2 r_1$ controls identifiability. A tangential target is not automatically singular: nonzero speed still supplies timing information. Once time is corrected, yaw can be estimated from a weighted circular resultant of predicted and observed bearings. This ordering avoids asking a single optimizer to explain the same discrepancy simultaneously as time, yaw and radial-speed bias.

7 Experimental Evaluation

7.1 Alignment protocol

The scalar race starts at 0° with a true offset of 1.2°. Each ten-frame block contains seven eligible frames with Gaussian observation standard deviation 0.3° and three other frames with 1.5° standard deviation. Both methods receive the same observations. We compare the original ten seeds and a 100-seed extension. The threshold is 0.1° and the evaluation ends at frame 400. A sequence that never enters the threshold is right-censored, not assigned a measured lock time of 400 frames. The original F1 ten-seed result is a mean first entry of 46.2 frames. Its 100-seed mean is 41.14 frames, with no censored fast sequences; all slow sequences remain censored. F2 and F3 remain censored in all 100 trials under this observation protocol. Therefore the manuscript does not attach "46 frames" to their parameters. Nor does it interpret a censored 400-frame baseline as a known 400-frame convergence time. At an assumed 20 Hz, the original mean corresponds to 2.31 s. File counts, recording duration and compute runtime are different quantities. No twenty-node wall-clock speedup or 360,000-frame production loss is inferred from this scalar experiment.

7.2 Timing and supporting estimators

The clock fixture recovers $\tau = 20$ ms and $b = 0.1$ m/s to numerical precision for its exact constant-velocity model. In 2,000 noisy draws per geometry, the single-target timing RMSE is 6.632 ms for the radial fixture and 3.799 ms for the tangential fixture. Position and velocity component noise have standard deviation 0.01 in their respective units; range noise is 0.01 m and radial-speed noise is 0.005 m/s. These controlled fixtures test the invariant, not a production synchronization service.

7.3 Resource-aware residual estimation

For a single complex FMCW tone, derotating a predicted beat frequency leaves a near-DC residual whose lag-one phase estimates frequency [3]. The available prototype uses 512 samples at 10 MHz, a 77 GHz carrier and chirp slope 5×10^{12} Hz/s. Its operation-count model gives 4,608 versus 1,024 complex products, a 4.5 ratio, but does not time an optimized DSP FFT. The recorded errors 0.4393, 0.1059 and 0.1075 m are mean absolute range error, despite an older print label calling them RMSE. The FFT interpolation implementation also requires a sign-correct denominator before serving as a strong accuracy baseline. We consequently use this study to motivate the compute envelope, not to claim superiority over a competitive spectral estimator.

8 Discussion and Conclusion

The practical lesson is to stop treating replay overlap as a substitute for an initialization model. Fast adaptation can reduce the first-entry delay, but only for a qualified observation stream and a gain budget large enough to remove the initial bias. Historical seeding can remove that adaptation phase in offline reconstruction, but only by making the provenance and evaluation mode explicit. The present evidence supports a controlled algorithm study, exact clock fixtures and an inspectable integration design. The production wrapper's gain readout and update-count cap are not a replacement for the buried calibration recursion. Persistent-lock tests, quality-qualified header extraction and a complete production replay comparison remain the next system-level evaluation. Those boundaries do not diminish the central result; they make clear which part of the replay problem has actually been solved.

A A Complete Error Budget for Fast Initialization

A.1 Three clocks in one experiment

There are three different indices in a replay experiment: wall-clock execution time, recording frame time and eligible-update count. The gain schedule is parameterized by the third. If a recording contains pauses, variable cadence or disqualified motion, an elapsed-time constant cannot be substituted for an update constant. A scheduler can process a minute of recording in a second of compute time without changing the estimator's sequence of accepted observations. For an approximately constant eligibility fraction p, the rough map $N_k \simeq pk$ makes the fast component decay on an elapsed-frame scale $1/(p\alpha)$. This is a useful design approximation, not an exact result for clustered eligibility. Two sequences with the same eligible fraction can have different first-entry behavior because early accepted observations carry larger gains and because rejected-frame noise still enters the model used in this paper.

A.2 Expected contraction under a random gate

If a_k is Bernoulli with probability p and the fast gain has not yet appreciably decayed, a local approximation gives $\mathbb{E}[g_k] \simeq K_s + pK_f$. For the actual accepted-count schedule, however, g_k and N_k are dependent. Replacing

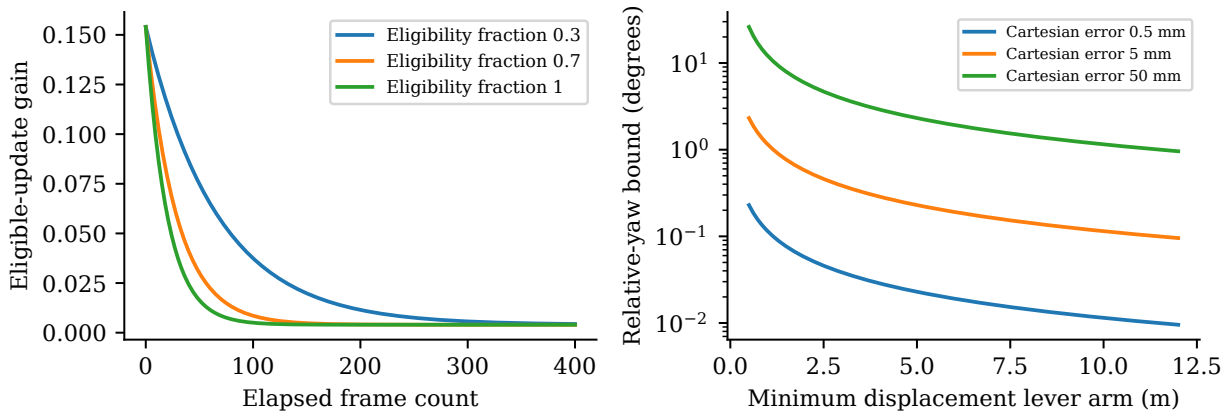


Figure 3: Left: elapsed-frame gain trajectories for different eligibility fractions illustrate why one warm-up duration is not portable. Right: deterministic relative-yaw error bounds tighten with displacement lever arm and degrade rapidly with Cartesian measurement error. These are analytical curves, not additional measured trials.

$\mathbb{E}[\prod(1-g_k)]$ by $\prod(1-\mathbb{E}[g_k])$ is not generally justified. Conditional analysis on a recorded eligibility sequence avoids this approximation altogether.

Proposition 1 (Conditional finite-horizon moments)**.** *For deterministic gains $g_0,\dots,g_{m-1}$ and mutually independent zero-mean noises independent of e_0, the error mean and variance are given by the product and variance expansion in the main text.*

Proof. Write $e_{k+1} = (1-g_k)e_k + g_k\eta_k$ and expand recursively. The coefficient multiplying e_0 is the full product, while the coefficient multiplying noise η_j is $g_j\prod_{i=j+1}^{m-1}(1-g_i)$. Independence removes cross terms when taking the variance. Taking expectations removes the zero-mean noise terms. □

The proposition is intentionally conditional. In a real stationary gate, eligibility is estimated from noisy observations, so selection may change the observation distribution. A gate that preferentially admits errors of one sign can bias the calibration even if the unselected noise is zero mean. Monte Carlo tests should therefore include the complete qualification mechanism, not just a periodic synthetic flag.

A.3 A threshold bound and its limits

Suppose the noiseless estimate begins with $|e_0| > \delta$. A sufficient first-entry condition follows by making the exponent in the contraction bound exceed $\log(|e_0|/\delta)$. The accumulated slow term grows as mK_s, while the fast budget saturates at $K_f e^{-\alpha}/(1-e^{-\alpha})$. This exposes why increasing α can reduce performance: a faster decay does not merely end adaptation sooner; it reduces the total available startup correction. For small α, the saturation budget is approximately $K_f/\alpha - K_f/2$. That approximation is accurate only when the discrete accepted-update convention is retained. Starting the exponent at zero rather than one adds one extra full fast-gain term and changes the finite-horizon experiment.

A.4 Persistent lock as a sequential event

Let $L_k = \mathbf{1}\{|e_k| < \delta\}$ and define persistent lock by $\prod_{i=k-H+1}^{k} L_i = 1$. Correlated estimator errors make its probability different from $\Pr(L_k = 1)^H$. A useful report includes the first-entry distribution, the time of first persistent lock, and the probability of leaving the tolerance after declaration. These statistics answer different

engineering questions: availability, stability and post-release reliability. The present scalar race records only first entry. Its 46.2-frame result cannot be relabeled as ten consecutive stable frames without rerunning a different scoring function. This is why the paper uses a precise event definition rather than the more attractive but ambiguous word “converged.”

B Warm-Up Cost and Shard Design

B.1 Overlap overhead

For M scored frames split among W workers and h preceding warm-up frames per worker after the first, total work is approximately $M + (W-1)h$. Ignoring I/O and imbalance, the redundant-work ratio is

$$\Omega = 1 + \frac{(W-1)h}{M}. \tag{13}$$

The wall-clock makespan is closer to $M/W + h$ than to total work. A shorter initialization horizon can therefore matter even when the redundant fraction looks modest. Conversely, when I/O or a serial decoder dominates, reducing filter warm-up may produce little end-to-end speedup. A publication should report both estimator delay and measured job runtime rather than infer one from the other.

B.2 Scored segments and state age

A preceding segment is useful only if its final state is valid at the first scored epoch. A header state obtained at a different timestamp must be propagated or explicitly interpreted as a mounting parameter that is constant over the interval. Dynamic clutter maps and target tracks do not share that constancy assumption. Seeding only the boresight angle cannot recreate all upstream and downstream state. State age also matters when the vehicle changes temperature or undergoes a mounting disturbance. Treating all calibration variables as static would hide precisely the cases where a warm-start strategy needs a fallback. The initialization contract should therefore identify which variables are copied, which are propagated and which are relearned.

B.3 Replay reproducibility versus online realism

An offline regression job may legitimately reproduce the final known calibration to compare downstream algorithms under identical front-end conditions. An online availability test may not. The two experiments are both useful but answer different questions. We recommend naming them “retrospectively seeded replay” and “causal cold-start replay,” so the evaluation mode appears in every table rather than only in a footnote.

C Seed Uncertainty and Decision Sensitivity

C.1 Fusion of a seed and an independent estimate

If a seed θ_s with variance P_s and an independent initial estimate θ_b with variance P_b are both valid, their Gaussian information fusion is

$$P_0 = (P_s^{-1} + P_b^{-1})^{-1}, \qquad \theta_0 = P_0(P_s^{-1}\theta_s + P_b^{-1}\theta_b). \tag{14}$$

Independence is essential. If both were derived from the same recording segment, treating them as independent double-counts information. A deterministic replay may simply select one declared source rather than pretend to have two independent measurements.

C.2 How angle uncertainty reaches association

For a point at range r and bearing ϑ, a small mounting error $\delta\theta$ causes

$$\delta\mathbf{p} \simeq r \begin{bmatrix} -\sin\vartheta \\ \cos\vartheta \end{bmatrix} \delta\theta. \tag{15}$$

The induced position covariance is $J_\theta P_\theta J_\theta^T$. At long range, even a small angle variance can dominate lateral uncertainty. Injecting a point seed without its uncertainty may therefore tighten an association gate just when the transformed detections remain uncertain.

C.3 A local decision-preservation radius

Suppose a decision compares scores $f_1(\theta)$ and $f_2(\theta)$ with nominal margin $m = f_2 - f_1 > 0$. If their local sensitivities are bounded by L_1 and L_2, the ordering is preserved whenever

$$|\delta\theta| < \frac{m}{L_1 + L_2}. \tag{16}$$

This does not certify a whole tracker unless all relevant comparisons and validity domains are included. It does, however, provide a principled alternative to choosing warm-up length only by file count. Near a tie, the admissible initialization error becomes small; far from a boundary, the same state error may be harmless.

D Deskew Derivation and Timestamp Error Propagation

D.1 Exact rigid-body interpretation

Let $T(t) \in SE(2)$ map a sensor-attached frame at time t into a fixed reference frame. A stationary world point observed as $\mathbf{p}(t)$ is transported to reference time t_0 by $T(t_0)^{-1}T(t)$. This expression defines the rotation and translation signs without ambiguity. Expanding the vehicle motion in $\Delta t = t_0 - t$ gives a second-order approximation whose signs depend on which of the two frame transforms is being approximated. The wrapper expression in the main text is a declared convention for a particular implementation. A verification fixture should first apply known pure translation, then known pure rotation, and finally combined motion. A sign that appears harmless under straight driving can reverse the correction during a turn.

D.2 Timestamp Jacobian

For the stated map, define $J = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$. Differentiation gives

$$\frac{\partial \mathbf{p}_c}{\partial \Delta t} = \omega J R(\omega\Delta t)\mathbf{p} - \mathbf{v}_e - \mathbf{a}_e\Delta t. \tag{17}$$

An offset variance σ_t^2 induces $J_t\sigma_t^2 J_t^T$ position covariance. Neglecting that term after deskewing assumes the timestamp correction is exact. The derivative also shows the long-range rotational lever arm: timing sensitivity grows with $|\omega|\,\|\mathbf{p}\|$.

D.3 Truncation and acceleration

For bounded jerk, the omitted translation is third order in Δt. For varying yaw rate, the omitted angle contains angular acceleration. These terms are small only relative to the intended position-error budget. A universal acceptable clock offset cannot be specified independently of speed, range and turn rate.

E Clock/Bias Identifiability and Error Propagation

E.1 Weighted multi-target extension

For more than two independent tracks, stack $y = A\mathbf{z} + \varepsilon$ with $\mathbf{z} = (\tau, b)^T$. If Σ_y is known and A is treated as fixed, the weighted least-squares solution is

$$\hat{\mathbf{z}} = (A^T \Sigma_y^{-1} A)^{-1} A^T \Sigma_y^{-1} y. \tag{18}$$

The information matrix must be nonsingular. When track velocity is noisy, the coefficients $a_i = \|\mathbf{v}_i\|^2$ are uncertain too, yielding an errors-in-variables problem. The simple covariance expression then understates uncertainty; a first-order augmented propagation or a joint likelihood is needed.

E.2 First-order single-target uncertainty

With $y = rw - \mathbf{d}^T \mathbf{v}$ and $a = \mathbf{v}^T \mathbf{v}$, the differential of $\tau = y/a$ is

$$d\tau = \frac{w\,dr + r\,dw - \mathbf{v}^T d\mathbf{d} - (\mathbf{d} + 2\tau\mathbf{v})^T d\mathbf{v}}{a}. \tag{19}$$

It reveals two different low-information regimes: small speed makes the denominator small, while large position or velocity uncertainty inflates the numerator. Tangential motion does not force a to vanish. This is consistent with the noisy tangential fixture, which is observable despite nearly zero instantaneous radial motion.

E.3 Constant-acceleration correction

For $\mathbf{x} = \mathbf{d} + \mathbf{v}\tau + \frac{1}{2}\mathbf{a}\tau^2$ and $\dot{\mathbf{x}} = \mathbf{v} + \mathbf{a}\tau$, direct expansion yields

$$rw - \mathbf{d}^T \mathbf{v} = (\|\mathbf{v}\|^2 + \mathbf{d}^T \mathbf{a})\tau \tag{20}$$

$$+ \tfrac{3}{2}\mathbf{v}^T \mathbf{a}\tau^2 + \tfrac{1}{2}\|\mathbf{a}\|^2 \tau^3. \tag{21}$$

The linear coefficient may become small even at nonzero speed if $\mathbf{d}^T \mathbf{a}$ cancels $\|\mathbf{v}\|^2$. An acceleration-aware solver should test that coefficient and the local derivative of the cubic, rather than rely solely on the constant-velocity speed test.

E.4 Circular yaw recovery

After timing correction, let ϑ_i be the measured local bearing and $\psi_i = \arg(\mathbf{d}_i + \tau\mathbf{v}_i)$ the predicted common-frame bearing. Then

$$\hat{\beta} = \arg \sum_i w_i \exp[j(\psi_i - \vartheta_i)]. \tag{22}$$

A small resultant magnitude signals weak or conflicting orientation evidence. Averaging angles arithmetically would mishandle the wrap at $\pm\pi$; the circular mean makes the geometry explicit. Incorrect pair correspondences can still create a strong wrong mode, so this estimator is not a replacement for data association.

F Ghost-Difference Calibration and Its Failure Budget

F.1 Relative yaw from a common displacement

Suppose two sensors observe the same physical direct-to-image displacement Δp_k in a planar complex representation. Their local versions are $D_{ik} = e^{-j\beta_i}\Delta p_k$ and $D_{jk} = e^{-j\beta_j}\Delta p_k$. Therefore

$$\widehat{\beta_j - \beta_i} = \arg \sum_k D_{ik}\overline{D_{jk}}. \tag{23}$$

The absolute target position cancels; relative yaw remains. Absolute yaw additionally requires baseline closure or an external orientation reference. With zero surveyed baseline, that absolute problem is unobservable even if the relative estimate is precise.

F.2 A deterministic relative-yaw certificate

If each Cartesian endpoint has norm error at most ε and each true displacement has magnitude at least $L_{\min}$, a conservative perturbation ratio is

$$\rho = \frac{4\varepsilon}{L_{\min}} + \left(\frac{2\varepsilon}{L_{\min}}\right)^2. \tag{24}$$

For $\rho < 1$, the associated resultant-angle error can be bounded by $\arcsin\rho$ under the shared-displacement assumptions. If the bound exceeds the release tolerance, the correct output is refusal, not a more optimistic point estimate. The analytical curves in Fig. 3 show why short displacement lever arms and realistic bearing-induced Cartesian errors can defeat a sub-tenth-degree goal.

F.3 Why single-scan geometry is only a coarse seed

At tens of metres, a few tenths of a degree of bearing noise correspond to decimetres of lateral uncertainty. Dividing that uncertainty by a six-metre displacement creates a degree-scale orientation error. Pooling true pairs can reduce independent noise, but structured wrong correspondences do not average away like zero-mean errors. A coarse geometry seed may accelerate a subsequent qualified filter; it should not be sold as an instant precision-calibration certificate.

G Residual Beat Kernel: Equations, Not a Source Dump

For sample interval T_s, predicted beat f_p, and complex samples x_n, the computational kernel is simply

$$z_n = x_n e^{-j2\pi f_p n T_s}, \tag{25}$$

$$C_1 = \sum_{n=1}^{M-1} z_n \overline{z_{n-1}}, \tag{26}$$

$$\hat{f}_b = f_p + \frac{\arg C_1}{2\pi T_s}. \tag{27}$$

The FMCW mapping includes both range and Doppler: $f_b = 2\mu r/c + 2v/\lambda$. A single chirp does not independently identify both unknowns. A track-conditioned range refinement must state where the velocity estimate comes from and propagate its uncertainty.

G.1 Unambiguous residual interval

The phase estimate is modulo 2π, so the residual is unambiguous only within $(-1/(2T_s), 1/(2T_s)]$. Derotation does not enlarge the information in a sampled tone; it centres the expected residual within the desired interval and can simplify subsequent arithmetic. An incorrect prediction that crosses the branch boundary can produce a discontinuous estimate.

G.2 Quantization and multiply counts

A common real/imaginary scale preserves phase more naturally than unrelated component scales. If vector quantization error is bounded by ε_q and the signal magnitude is $A > \varepsilon_q$, a sample-angle perturbation is bounded by $\arcsin(\varepsilon_q/A)$. Accumulation, cancellation and low-amplitude samples require their own analysis. Calling a floating-point fake-quantized experiment "no-FPU INT8 firmware" would overstate what was tested. The 4.5 product-count ratio follows from the chosen models $M\log_2 M$ and $2M$ at $M = 512$. An optimized radix-two FFT has a different exact arithmetic count, and oscillator generation has a cost too. Hardware profiling, rather than the algebraic ratio, decides whether the residual kernel is worthwhile on a particular DSP.

H Reproducibility and Publication Checklist

All numerical comparisons in the main text identify initialization, observation noise, eligibility, gain configuration, seed range and scoring event. The timing fixtures specify physical units and the source of each independent track. The real-data calibration example establishes that transport fields exist; it is not mixed into the scalar estimator's synthetic accuracy table. No full source files or repository diffs are part of this public manuscript. A reviewer needs an algorithm, a protocol and sufficient equations to reconstruct the experiment, not an internal working-tree archive. The original source can remain an internal reproducibility artifact or be released separately under an appropriate project decision. The current study does not establish the wall-clock performance of a twenty-worker deployment. A complete system report would include the fraction of qualified frames per drive, persistent-lock distributions, seed rejection rates, scored output differences and actual makespan. Those measurements would test whether the mathematical initialization policy delivers the expected practical benefit.

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